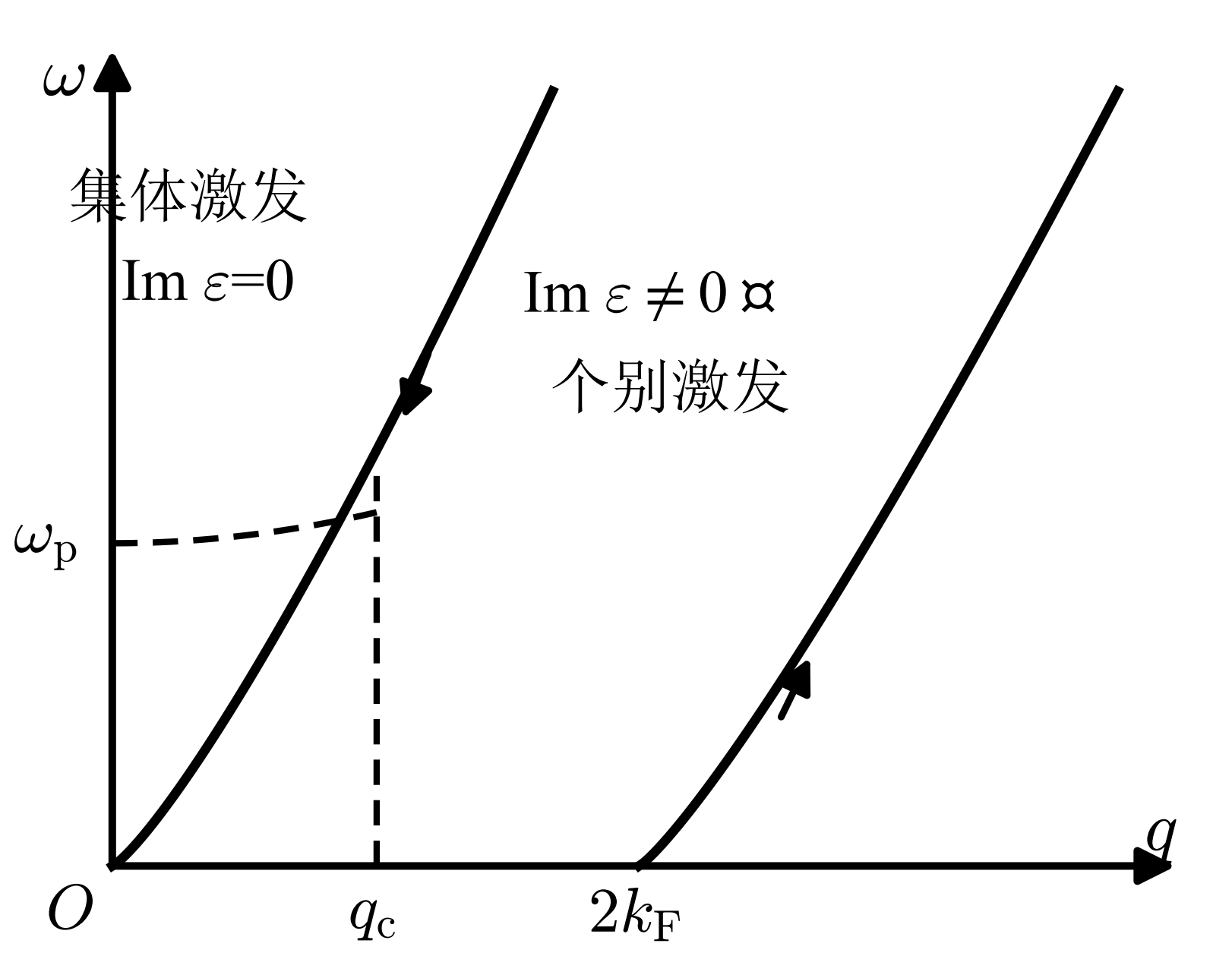
ω
集体激发
Im ε=0
Im ε ≠ 0 ¤
个别激发
ω_p
O
q_c
2k_F
q